

K2-29b

R-2336

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_250904 · created 2026-08-18T23:48:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460922.8766 +/- 0.0084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.047
Transit depth (Rp/Rs)^2	2.39 +/- 1.47 [%]
Orbital Inclination (inc)	84.38 +/- 2.44
Airmass coefficient 1 (a1)	1.152 +/- 0.084
Airmass coefficient 2 (a2)	-0.104 +/- 0.031
Scatter in the residuals of the lightcurve fit is	7.19 %
Best Comparison Star	#1 - [391, 242]
Optimal Method	PSF photometry
Transit Duration (day)	0.049 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.155 ± 0.047 vs 0.1419 (deviation 0.16σ)
Duration fit vs published	0.049 d vs 0.0925 d (deviation 0.68σ)
Mid-time O-C	-21.0 min
Depth SNR	1.9
Residual scatter	7.19 %

Light curve

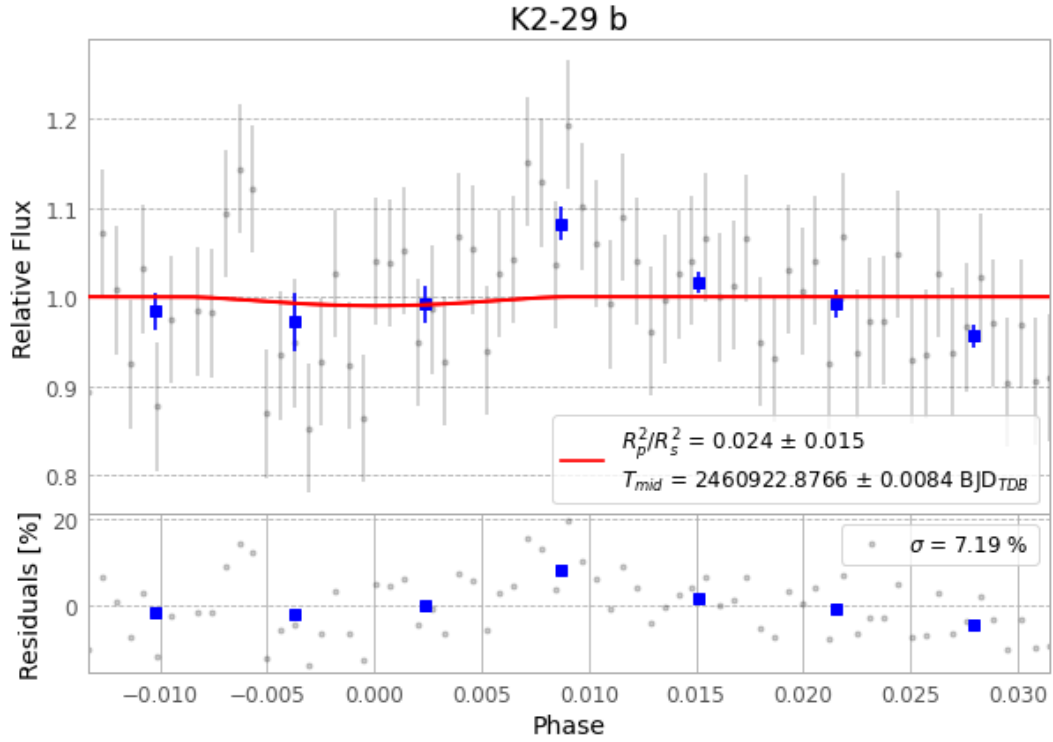


Figure 1 — FinalLightCurve_K2-29 b_2025-09-04.png (SHA-256 ca10f28eca4f5890...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 436], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 28b2a07a29f19568...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

71 FITS frames (47 MB), K2-29250904075716.FITS → K2-29250904112715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-250904.log (9123bf47f2d1...), FinalLightCurve_K2-29 b_2025-09-04.png (ca10f28eca4f...), FinalLightCurve_K2-29 b_2025-09-04.csv (8ff6fc59139d...)

Compute resources

Wall time 179.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:48Z.